INCH-POUND

MS27481F
12 July 2002
SUPERSEDING
MS27481E
13 August 1974

DETAIL SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, ELECTRICAL, JAM NUT MOUNTING, CRIMP TYPE, BAYONET COUPLING, SERIES II

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-38999.

Inactive for new design.

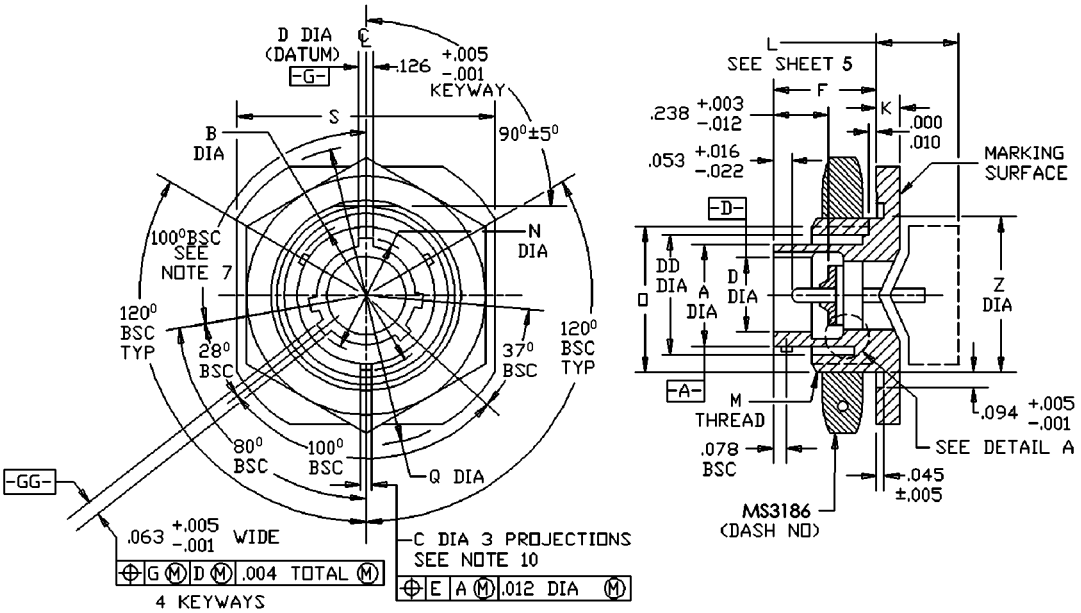
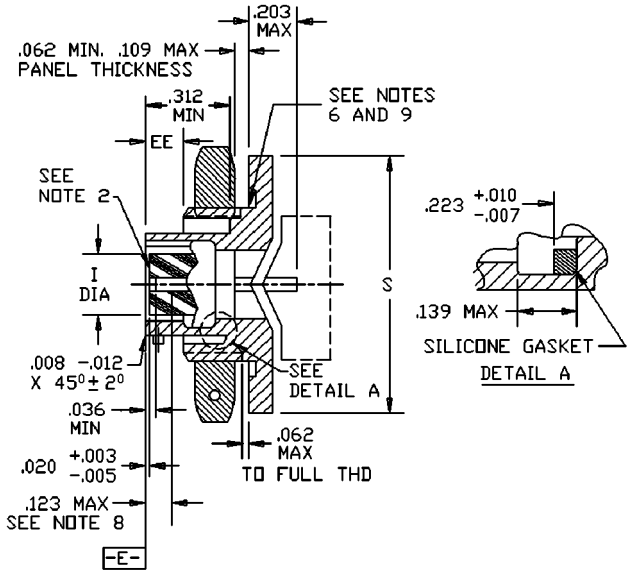


FIGURE 1. Receptacle, classes E and T.

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Shell size	A dia +.001 -.005	B dia +.000 -.016 over proj	C dia +.006 -.002	D dia +.005 -.001	F ±.005	I dia ±.005	K +.011 -.010	M thd class 2A Plated	N dia +.005 -.006
8	.473	.563		.362		.285		.875 – 20 UNEF	.412
10	.590	.680		.490		.413		1.000 – 20 UNEF	.540
12	.750	.859	.078	.607	.438	.527		1.125 – 18 UNEF	.689
14	.875	.984		.732		.652		1.250 – 18 UNEF	.814
16	1.000	1.103		.857		.777	.094	1.375 – 18 UNEF	.939
18	1.125	1.233		.962		.866		1.500 – 18 UNEF	1.039
20	1.250	1.358		1.087		.991		1.625 – 18 UNEF	1.164
22	1.375	1.483	.125	1.212	.464	1.116		1.750 – 18 UNS	1.289
24	1.500	1.610		1.337		1.241		1.875 – 16 UN	1.414

FIGURE 1. Receptacle, classes E and T – Continued.

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Shell size	O FLAT +.001 -.006	Q dia ±.016	S ±.016	Z dia +.006 -.005	CC	DD min dia	EE +.005 -.000	Recommended packing ("O" ring) see note 9	For insert arrangement, see MS drawing
8	.817	1.375	1.250	.965	.004	.678		AS3578 – 022	27343, 27516
10	.941	1.500	1.375	1.090		.780		AS3578 – 024	27344, 27517
12	1.065	1.625	1.500	1.215		.963	.140	AS3578 – 026	27345, 27518
14	1.190	1.750	1.625	1.340		1.088		AS3578 – 028	27346, 27519
16	1.320	1.938	1.781	1.477	.010	1.222		AS3578 – 029	27347, 27520
18	1.440	2.016	1.890	1.602		1.333		AS3578 – 030	27348, 27521
20	1.565	2.141	2.016	1.727		1.458		AS3578 – 031	27349, 27522
22	1.690	2.265	2.140	1.852		1.583	.166	AS3578 – 032	27350, 27523
24	1.815	2.390	2.265	1.977		1.708		AS3578 – 033	27351, 27524

NOTES:

1. Dimensions are in inches.
2. Insert front surface shall be flat within .005 T.I.R.
3. Diameters A, D, M, Q and Z shall be concentric within .005 T.I.R.
4. Diameters D and I shall be concentric within .005 T.I.R at MMC.
5. Diameter A with respect to diameter B, and diameter D with respect to diameter N, shall be concentric at MMC.
6. Sides of groove may have 5° max taper.
7. Normal keyway position. For other keyway positions see MIL-DTL-38999.
8. The point at which a gage pin, having the same basic diameter as the mating contact and a square face, touches socket contact spring.
9. Recommended packing is for dimensions only. Material is optional.
10. The gauge features for GG shall be CC smaller than their MMC size at basic location.

FIGURE 1. Receptacle, classes E and T – Continued.

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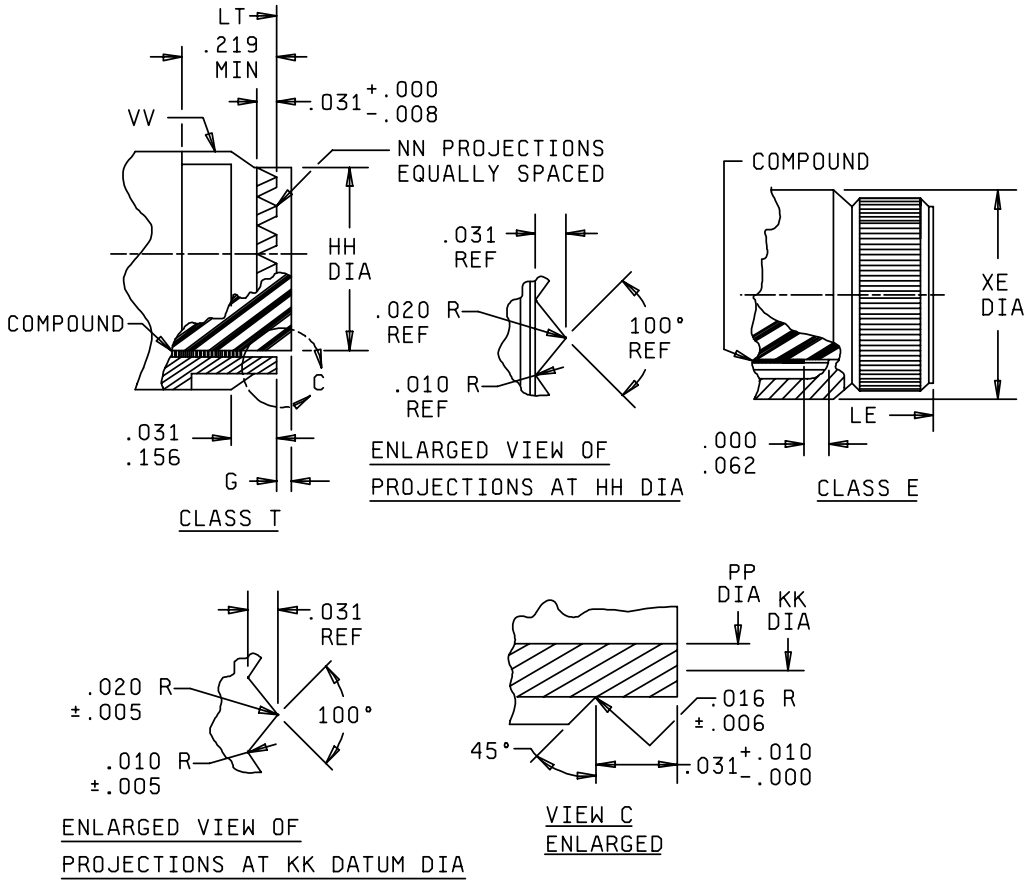


FIGURE 1. Receptacle, classes E and T – Continued.

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Shell size	HH dia +.000 -.005	KK DATUM dia	NN	PP dia ± .003
8	.387	.366	12	.344
10	.515	.487	16	.472
12	.628	.609	20	.586
14	.754	.731	24	.711
16	.879	.853	28	.836
18	.985	.974	32	.942
20	1.110	1.097	36	1.067
22	1.235	1.218	40	1.192
24	1.360	1.340	44	1.317

Shell size	Nut MS3186-	G insert projection class T only	LE max length	LT max length	XE max dia	VV thread UNEF-2A MOD (plated)	
						Size	Major dia MOD
8	8C	.120 ± .030	.516	.344	.594	7/16 – 28	.421 – .417
10	10C				.719	9/16 – 24	.542 – .538
12	12C				.844	11/16 – 24	.667 – .663
14	14C				.969	13/16 – 20	.791 – .787
16	16C				1.094	15/16 – 20	.916 – .912
18	18C				1.219	1 1/16 – 18	1.034 – 1.030
20	20C		.484	.312	1.344	1 3/16 – 18	1.158 – 1.154
22	22C				1.469	1 5/16 – 18	1.283 – 1.279
24	24C	.090 ± .050		.375	1.594	1 7/16 – 18	1.408 – 1.404

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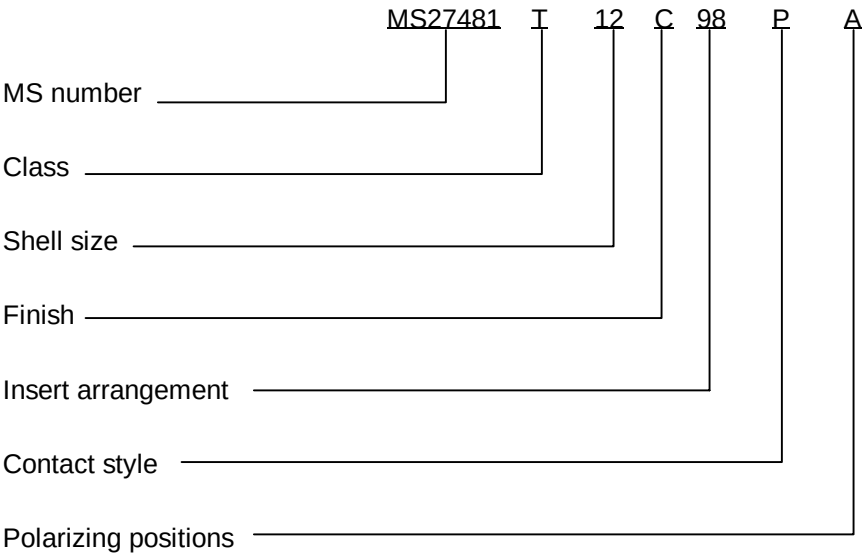
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REQUIREMENTS:

Dimensions and configuration: See figure 1. Interface dimensions shall conform to MIL-DTL-38999. This receptacle mates with MS27473, MS27480, MS27484 and MS27500.

Insert arrangements: See MIL-STD-1560.

Part or Identifying Number (PIN) example:



no letter is required for normal.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 11
DLA – CC

Preparing activity:
DLA – CC

(Project 5935–4375)

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19, 99
DLA – IS

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